



Khalil Shaikh, Co-Founder

farmer-friendly insects. The system has multiple versions tailored to specific uses. The lifespan of these systems is over six years without the need for any maintenance, and the cost of ownership to the buyer is between ₹37,800 and ₹52,300, depending on the variant.

BraveHawk. This AI-enabled technology utilises multiple sensors to identify different types of insects, explicitly targeting herbivorous pests that damage crops while sparing beneficial species. BraveHawk comes in two versions, one with solar and wind-powered capabilities and another with grid-powered capability: one suitable for open farm areas covering up to 2.5 acres



Shahnaz Shaikh, Co-Founder

and another designed for protected environments like greenhouses. Priced at ₹350,000, the device for open fields is powered by wind and solar energy, eliminating the need for electricity. Each unit is built to last over seven years with minimal maintenance.

eBionic. This bioelectronics technology focuses on exterminating specific targeted insect species, including various fruit flies, fruit-piercing moths, palm weevils, tea mosquito bugs, and cabbage butterflies. Priced at ₹350,000, the eBionic devices powered by wind and solar energy are narrow-spectrum and highly specific in their targeting, making them ideal for precise pest

management. The lifespan of these devices is over seven years without the need for any maintenance, and the cost of ownership to the buyer is around ₹419,000.

In addition to these technological solutions, the company offers non-tech-related support to farmers, enhancing their ability to manage farms sustainably. Future plans include expanding their portfolio with more broad-spectrum systems and deploying a portable laboratory to aid farmers in self-examining various farm parameters, such as soil, water, plant tissues, and fertigation solutions. Looking towards the future, the company is committed to expanding its impact by making these innovative tools accessible to small-scale farmers, a group that typically faces barriers to adopting advanced technologies due to cost constraints. To achieve this, they are in the process of collaborating with various non-banking financial companies (NBFCs) to develop a financing programme.

Overall, these products have significantly contributed to improving the quality and yield of crops by eliminating the need for toxic pesticides and insecticides.

—Aryaman Raghuvanshi

3 WIRELESS LIGHT-BASED COMMUNICATION WITH VELMENNI'S TECHNOLOGY

Velmenni is at the forefront of wireless communication innovation with its pioneering light-based technologies, Li-Fi and LC Link.

Founded by Deepak Solanki in October 2014 in New Delhi, Velmenni has developed two key solutions: an indoor Li-Fi system as a Wi-Fi alternative, and an outdoor LC Link for long-distance communication, offering a replacement for fibre cables and microwave RF technology in connecting mobile towers.

LC Link delivers a high-speed, wireless solution that replaces traditional fibre cables by using light to connect mobile towers. This advancement dramatically enhances network quality and the user experience.

Li-Fi, a highly energy-efficient technology, uses significantly less power than Wi-Fi for data

transmission, consuming under 10 watts per light, Deepak explains. By integrating Li-Fi into office lighting, a single light, such as a 2x2 panel, can support up to 16 users, functioning similarly to a Wi-Fi router.

Li-Fi supports point-to-multiple communication, allowing numerous users to connect to a single light